Tutorial 5

1 Taxes

There are two assets A and B. Asset A is not taxed whereas asset B is taxed on its returns. Let r be the interest rate and τ be the tax rate. Neither asset pays out any money in period 0.

1. Calculate rates of return $p_1^A/p_0^A - 1$ and $p_1^B/p_0^B - 1$ (where p_t^k is the price of asset k in period t).

2. Now, suppose asset A pays out y_0 in period 0. This dividend is un-taxed. Calculate $p_1^A/p_0^A - 1$.

3. Now, suppose asset B pays out y_0 in period 0. This dividend is un-taxed. Calculate $p_1^B/p_0^B - 1$.

2 Contingent Consumption and Assets

Suppose there are two equally likely states: $S = 2$ and $\pi = (\frac{1}{2}, \frac{1}{2})$.

1. What is the expected value of $x = (5, 3)$ and $x' = (2, 8)$?

2. Suppose can buy contingent consumption directly at prices $p = (1, 4)$. Suppose the consumer can spend 6. Can the consumer afford $(2, 1)$? What about $(1, 2)$?

3. Now, suppose you cannot buy contingent consumption directly. Suppose there are two assets: a risk free asset $a_f = (1, 1)$ and a risky asset $a_r = (4, 1)$. The risk free asset costs 1 and the risky asset costs 2. Suppose you cannot short and suppose the consumer has 3 to spend. Can the consumer afford the contingent consumption bundle $(5, 2)$? What about $(4, 4)$? What about $(2, 3)$?

4. Now, suppose that the assets can be held short. Find the state price vector $p = (p_1, p_2)$. That is, p is the price of contingent consumption so that, for any $\theta = (\theta_f, \theta_r)$ the price

of θ is the same as the price of the contingent consumption delivered by θ:

$$\theta_f + 2\theta_r = p \cdot (\theta_f a_f + \theta_r a_r), \qquad \text{for all } \theta_f, \theta_r \tag{1}$$

The left side of (1) is the price of the portfolio θ and the right is the price of the contingent consumption bundle delivered by the portfolio θ. Hint: take $(\theta_f, \theta_r) = (1, 0)$ and $(\theta_f, \theta_r) = (0, 1)$ in (1) and solve for (p_1, p_2). This is two equations and two unknowns.

5. Suppose you want to buy contingent consumption $(6, 3)$. How much of each asset do you need to purchase? How much does this cost? How much does $(6, 3)$ cost according to the state prices $p = (p_1, p_2)$ you found in the previous question?

3 Contingent Consumption and Insurance

Suppose there are two states where the first state has probability $\frac{1}{3}$: $S = 2$ and $\pi = (\frac{1}{3}, \frac{2}{3})$.

1. What is the expected value of $x = (6, 3)$ and $x' = (9, 12)$.

2. Suppose the consumer has utility function $U(x_1, x_2) = \frac{1}{3}u(x_1) + \frac{2}{3}u(x_2)$ where $u(x) = \sqrt{x}$ and suppose the consumer is endowed with $\omega = (0, 9)$. Does the consumer prefer his endowment or its expected value: $x = (6, 6)$?

3. Suppose the consumer can buy insurance. Specifically, can buy consumption in state 1 at the cost of one unit of consumption in state 2. That is, insurance is an asset $(1, -1)$. Find the safe bundle (k, k) (for some value of k) that the consumer can afford. Does the consumer prefer this bundle or the endowment?

4. Draw a picture of all the contingent consumption bundles which are affordable (i.e. draw the budget line).

5. Can you find a price vector $p = (p_1, p_2)$ so that the price of a contingent consumption x is the same for every affordable contingent consumption bundle:

$$p \cdot \omega = p \cdot (\omega_1 + \theta, \omega_2 - \theta), \qquad \text{for all } \theta$$

Note that there is more than one price vector p which satisfies this equality.

4 EU maximization

Suppose there are two states ($S = 2$) and that $\pi = (\frac{1}{3}, \frac{2}{3})$. Suppose utility takes the form

$$U(x_1, x_2) = \tfrac{1}{3}u(x_1) + \tfrac{2}{3}u(x_2)$$

where $u(x) = \sqrt{x}$. We note that $u'(x) = x^{-1/2}$.

1. Find $\text{MRS}_{1,2}$?
2. Find the demand function for the consumer. That is, suppose you can buy contingent consumption with price vector $p = (p_1, p_2)$ and expenditure level m. Find how much contingent consumption is demanded for each price vector p and each m.
3. Suppose the consumer has endowment $\omega = (0, 9)$ and faces prices $p = (2, 1)$. What contingent consumption bundle will the consumer buy?
4. Continue to assume that the consumer is endowed with $(0, 9)$ but now the consumer cannot purchase contingent consumption directly. Instead he can only buy insurance. Each unit of insurance costs two units of consumption from state 2 and delivers 1 unit of consumption in state 1. Draw the budget line of all consumption bundles which the consumer can afford. How much contingent consumption will the consumer demand? Hint: use the answer from the previous question.